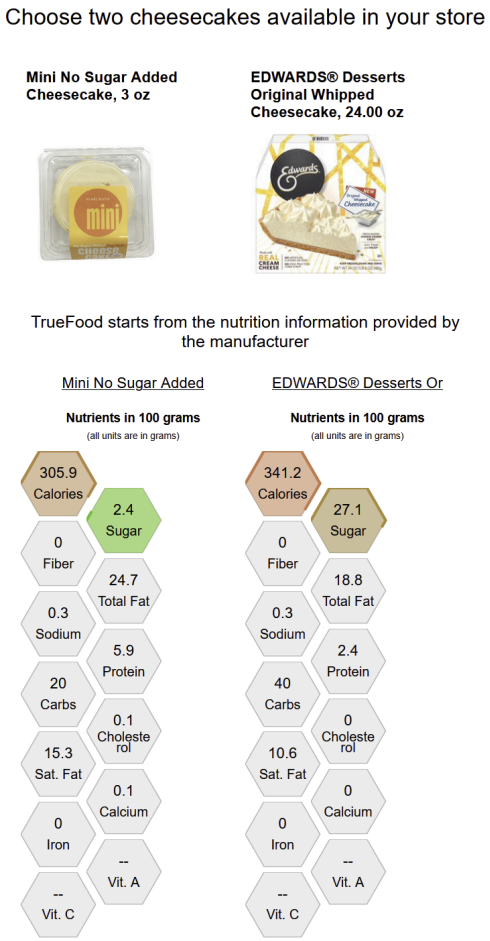
Overview and Motivation

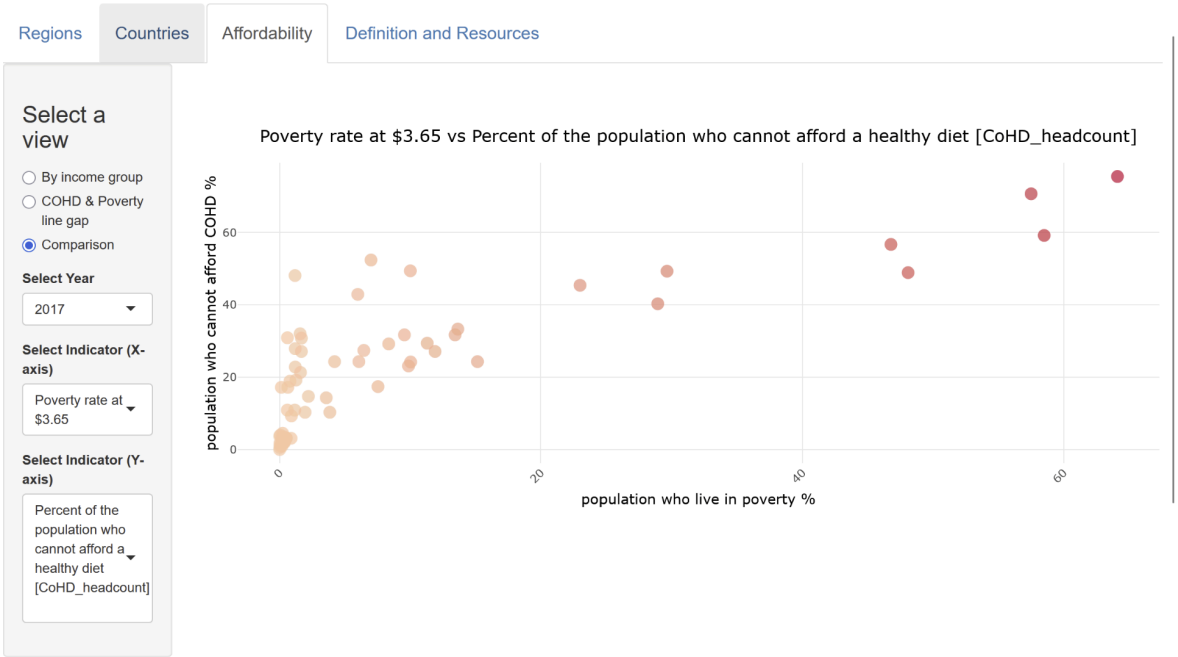
I chose this topic, because when shopping I do look for healthier alternatives to some products I normally get in stores when I can. However, I feel like some companies are a bit deceptive in the marketing for them. Like increasing pricing drastically for "more healthier" products like a "health tax". For instance, adding more protein and raising prices or healthier drinks that contain less sugar than their alternatives.

What I would like to learn from this project is how nutrients may impact pricings by attempting to identify any patterns that exist between factors such as pricing, calories, nutrition. Different pricings between different stores and food categories will have to be taken into account as well.

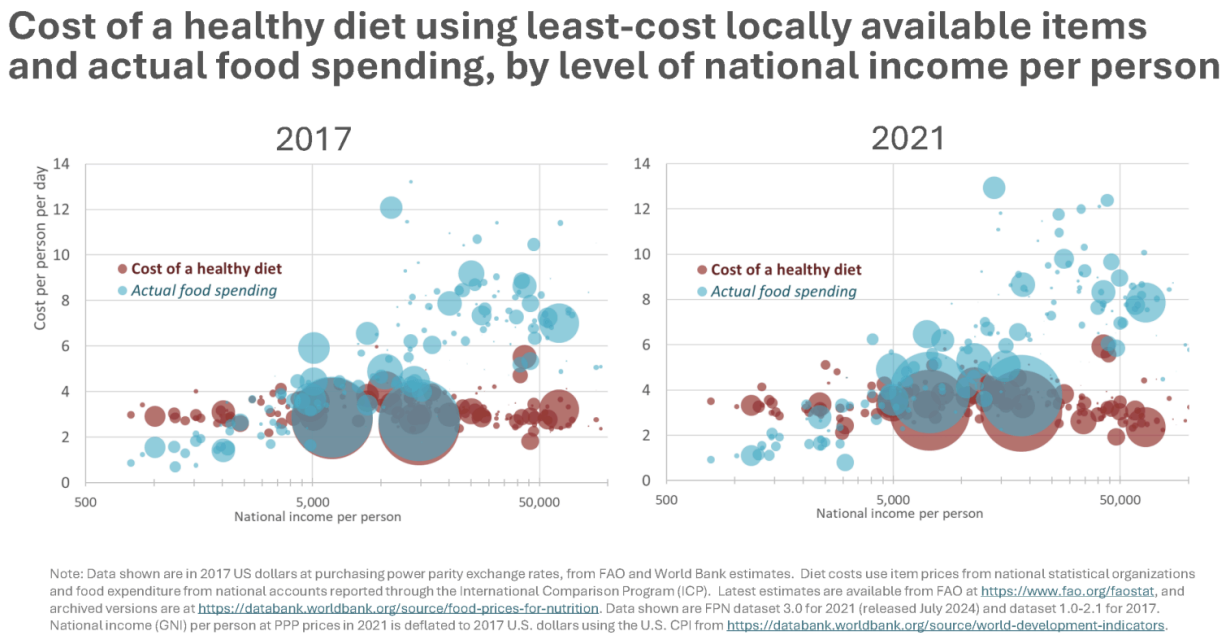
Related Work



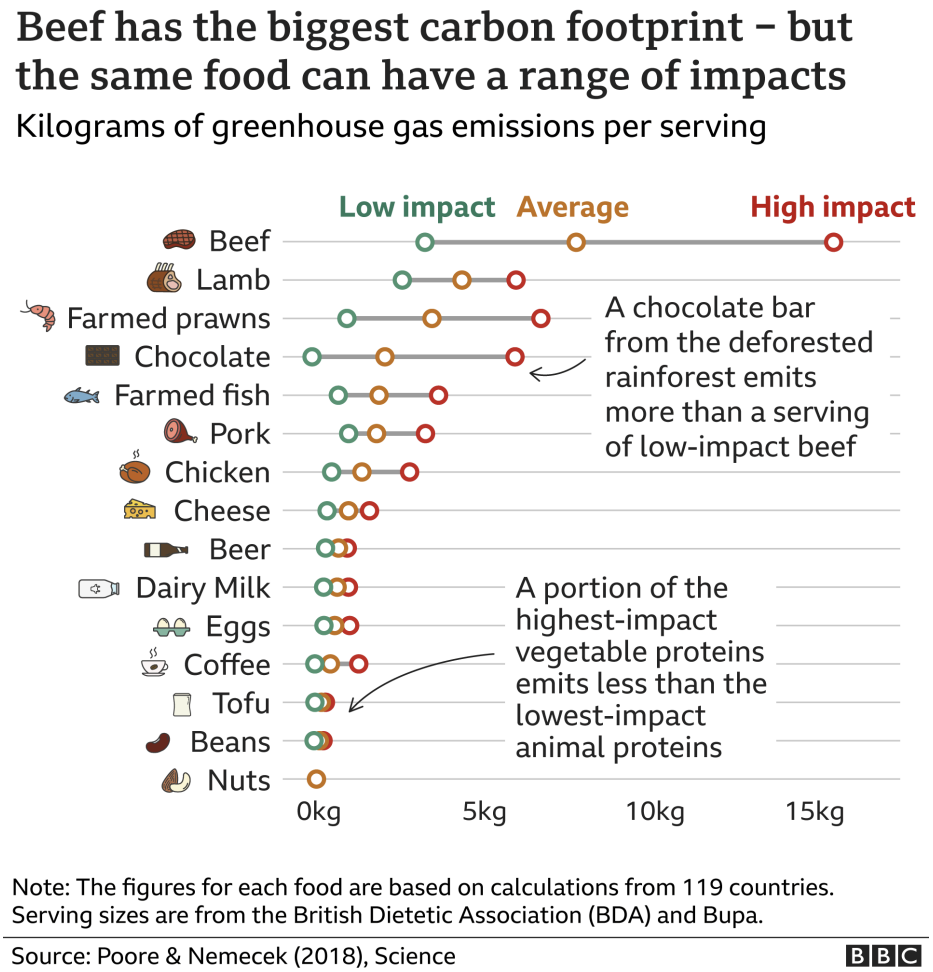
The original website and data visualization the dataset was used for.
https://www.truefood.tech/?store=all



This is more of something similar that I liked for an interface for the initial conceptualization of the scatterplot.
https://www.foodsecurityportal.org/node/3597



Something interesting that I have seen before and from a prior lab where points are scaled based on size to convey information, in this case, pricing.
https://sites.tufts.edu/foodpricesfornutrition/



Ideas for conveying potential ranges for general categories. A box and whisker plot could work similarly too.

https://www.bbc.com/news/science-environment-46459714

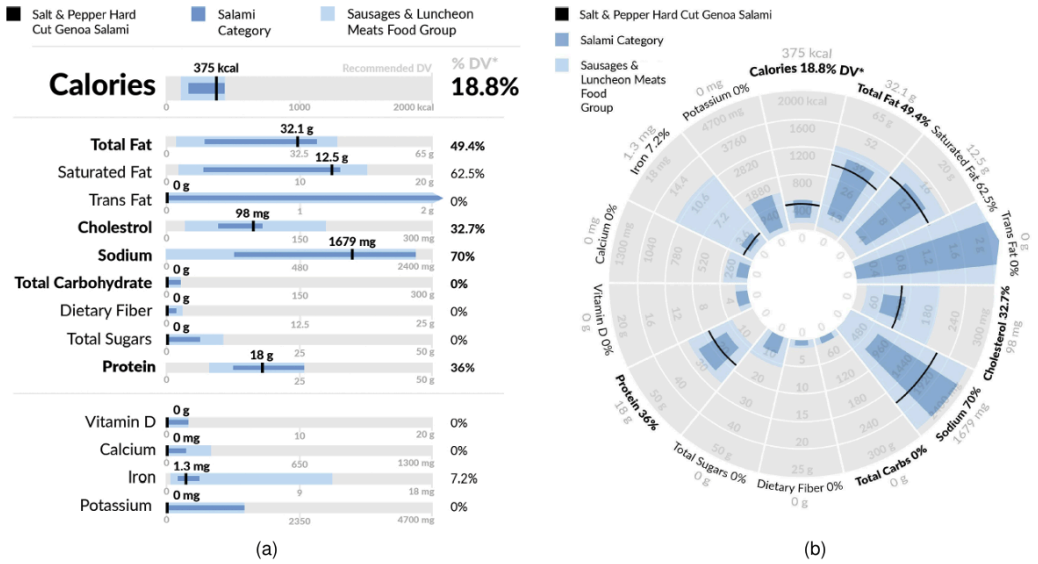
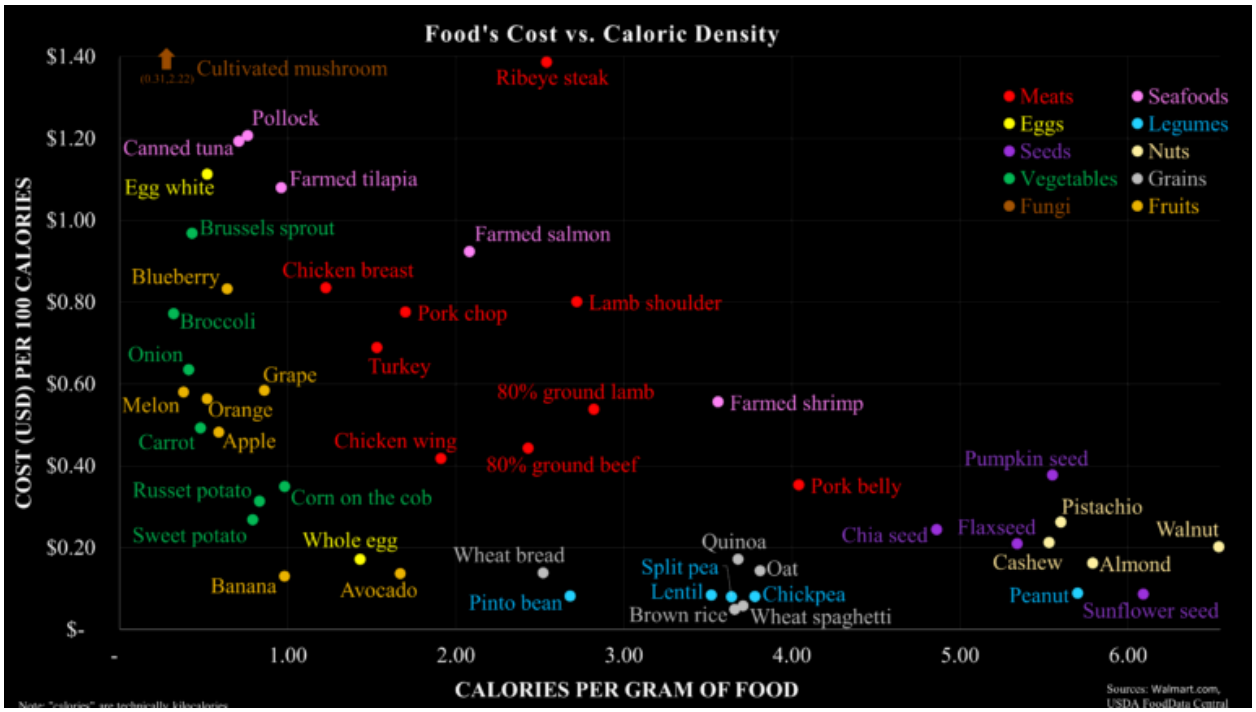


Figure 1: Examples of our linear (a) and radial (b) visual nutrition facts label designs based on the bullet graph. Both labels show the nutritional content of the food 'Salt & Pepper Hard Cut Genoa Salami' as well as the range of nutrient levels for all foods in the 'Salami' category (dark blue) and the 'Sausages & Luncheon Meat' food group (light blue).

Bar chart and radial chart examples or ideas for displaying or comparing values for nutrients. Radial might not be the best due to the usage of angles which makes it harder to convey the differences between bars.
https://www3.cs.stonybrook.edu/~mueller/papers/VisComm%202020.pdf



Another example for scatterplots with something a bit closer to my topic idea.
https://infographicsite.com/infographic/food-cost-versus-caloric-density-analysis/

Questions

Does the content or lack of content of any nutrients affect the pricings of products?

- Protein, Fat, Carbs, Sugar, Fiber. Calcium, Iron, Sodium, Vitamin C, Cholesterol, Fatty Acids, Vitamin A

Does retail at different stores affect pricing on nutrition, or is it irrelevant?

Are there any categories of products that are majorly affected by pricings based on nutrients?

Data

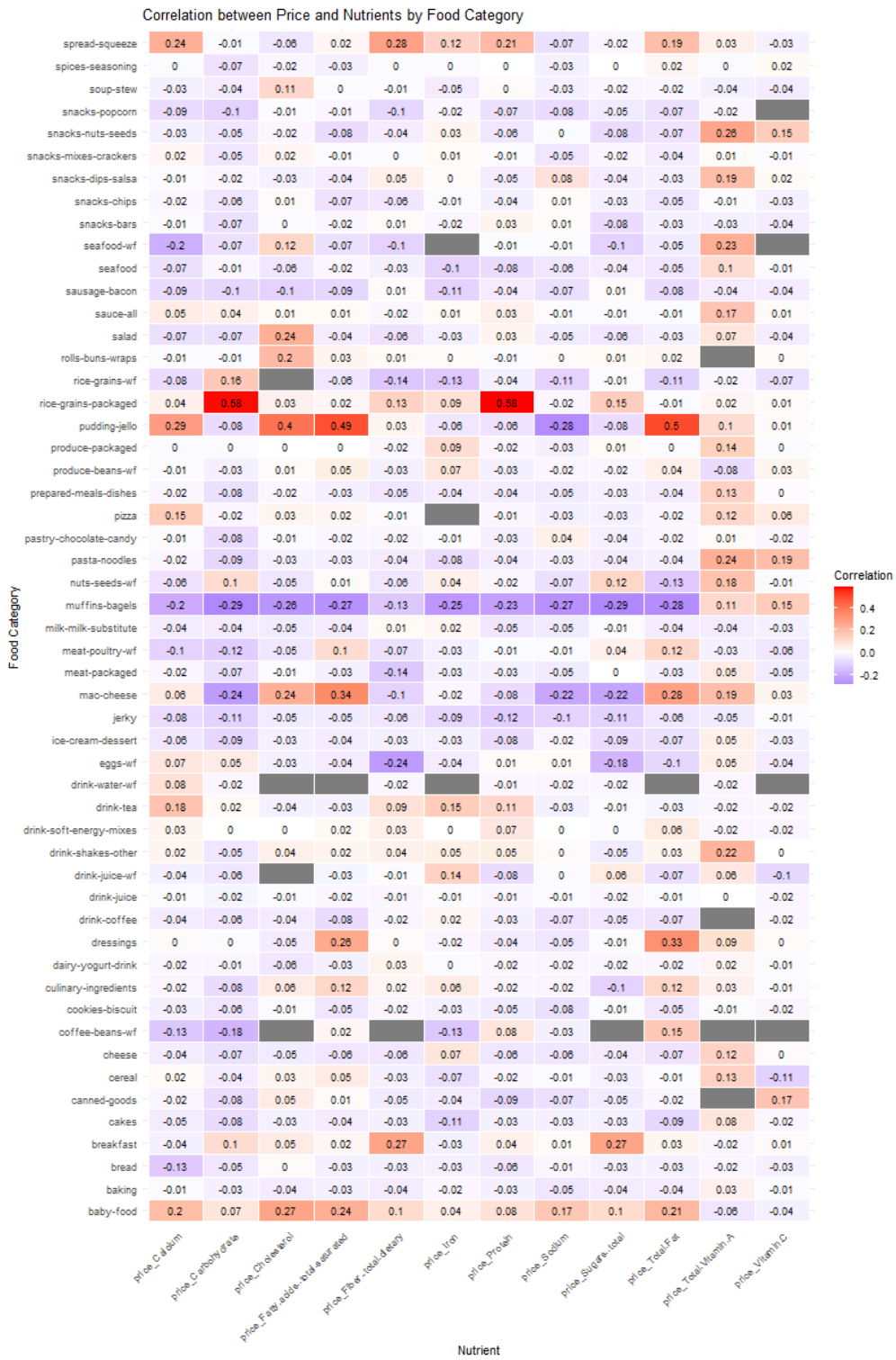
The data used is from this dataset which has been already scrapped from product information online from Walmart, Target, and Whole Foods's websites by Babak Ravandi et al. (https://www.nature.com/articles/s43016-024-01095-7). About 50,000+ food products were scrapped for data, which included nutritional information and their ingredient lists.

https://www.data-is-plural.com/archive/2025-01-29-edition/
https://github.com/Barabasi-Lab/GroceryDB/tree/main/data

I cleaned up the dataset by altering the csv in Google Sheets to make the data more understandable and fix potential problems. This was by primarily cleaning up empty cells by filling them in with data or NaN. I ended up discarding all products that didn't have pricing as since the goal was focused on price, having products that lacked that data point was not deemed needed. I also rounded a bunch of the numbers in the nutrients and pricing per calories to two decimal places to make it easier to comprehend. Finally, I fixed text issues in product titles where special characters were being replaced by ASCII codes.

Exploratory Data Analysis

Heatmaps for Correlation and Regression for All Stores



Regression Slopes of Price on Nutrients by Food Category

Food Category	Calcium	Carbohydrate	Cholesterol	Fatty acids-total saturated	Fiber-total dietary	Iron	Protein	Sodium	Sugars-total	Total Fat	Total Vitamin A	Vitamin C
spread-squeeze	7.98	0	-10.89	0.02	0.24	252.33	0.07	-1.22	0	0.03	104.56	-7.01
spices-seasoning	-0.05	0	-3.19	0	0	-1.29	0	0	0	0.01	-0.19	1.99
soup-stew	-0.11	0	0.64	0	0	-4.43	0	-0.01	0	0	-0.22	-0.23
snacks-popcorn	-16.13	-0.01	-0.01	0	-0.06	-42.31	-0.03	-0.38	-0.01	-0.01	-263.01	
snacks-nuts-seeds	-0.32	0	-3.04	-0.04	0	23.34	0	0	-0.01	0	595.93	64.76
snacks-mixes-crackers	0.27	0	1.95	-0.01	0	6.91	0	-0.21	0	-0.01	1.87	-0.48
snacks-dips-salsa	-0.1	0	-0.8	0	0.01	4.17	0	0.08	0	0	26.04	2.05
snacks-chips	-1.15	0	0.11	-0.03	-0.04	-1.6	-0.01	0	-0.02	0	-16.45	-4.39
snacks-bars	-0.13	0	-0.91	-0.01	0	-18.59	0.01	0	-0.01	0	-22.2	-23.92
seafood-wf	-22.71	-0.17	22.26	-0.46	-1.04		-0.02	0	-0.52	-0.07	202.29	
seafood	-0.54	0	-0.14	0	-0.02	-47.8	0	-0.01	-0.01	0	415.76	-7.63
sausage-bacon	-0.67	-0.01	-0.87	-0.01	0.01	-54.04	0	-0.06	0	0	-79.76	-2.11
sauce-all	0.13	0	0.04	0	0	5.87	0	0	0	0	166.27	1.23
salad	-0.37	0	11.52	-0.01	-0.01	-6.92	0	-0.03	0	0	27.22	-2.8
rolls-buns-wraps	-0.02	0	18.84	0.01	0	0.44	0	0	0	0		0.15
rice-grains-wf	-7.52	0.05		-10.81	-0.09	-151.42	-0.03	-57.56	-0.03	-0.25	-67.68	-51.92
rice-grains-packaged	3.03	0	40.1	0.06	0.1	87.28	0.01	-0.11	0.23	-0.01	54.71	1.79
pudding-jello	6.02	0	22.63	0.23	0.04	-100.11	-0.01	-0.79	0	0.15	53.65	8.58
produce-packaged	-0.02	0	-0.12	0	0	48.36	0	0	0	0	11.71	-0.09
produce-beans-wf	-0.2	0	2.55	0.07	0	46.3	0	0	0	0.02	-2.67	1.24
prepared-meals-dishes	-0.06	0	-0.14	0	0	-11.47	0	0	0	0	165.37	-0.06
pizza	4.24	0	0.48	0	0		0	-0.02	0	0	261.87	46.13
pastry-chocolate-candy	-0.08	0	-0.11	0	0	-2.32	0	0.01	0	0	49	-1.06
pasta-noodles	-0.04	0	-0.01	0	0	-18.23	0	-0.08	0	0	664.22	331.18
nuts-seeds-wf	-3.04	0.02	-51.07	0.01	-0.04	20.76	-0.01	-0.6	0.03	-0.04	8.06	-11.41
muffins-bagels	-0.14	0	-0.98	0	-0.01	-17.3	-0.01	-0.07	0	0	162.4	315.8
milk-milk-substitute	-0.02	0	-0.36	0	0	11.18	0	-0.04	0	0	-25.81	-2.54
meat-poultry-wf	-39.95	-0.12	-4.58	0.1	-0.39	-29.58	0	-0.09	0.32	0.07	-7.39	-117.36
meat-packaged	-0.29	-0.02	-0.29	0	-0.37	-61.35	0	-0.07	0	0	74.48	-50.29
mac-cheese	0.82	-0.02	39.22	0.38	-0.11	-48.7	-0.03	-1.62	-0.13	0.14	239.14	12.22
jerky	-2.6	0	-0.09	0	-0.15	-20.38	0	-0.04	0	0	-120.55	-3.11
ice-cream-dessert	-0.1	0	0	0	-0.01	-11.74	0	-0.01	0	0	33.08	-0.86
eggs-wf	0.03	0	-0.03	-0.01	-1.67	-6.61	0	0.03	-0.09	0	36.4	-70.73
drink-water-wf	11.5	0			-0.01		-5.32	-1.46	-0.01		-20.72	
drink-tea	22.08	0	-64.08	-0.06	0.07	700.28	0.27	-1.48	0	-0.04	-219.46	-1.76
drink-soft-energy-mixes	0.3	0	-0.15	0	0.04	2.11	0.05	0	0	0.03	-174.95	-0.41
drink-shakes-other	0.08	0	7.95	0.02	0.01	26.68	0	0.02	0	0.01	526.09	-0.01
drink-juice-wf	-1.23	0		-0.23	-0.01	120.02	-0.09	-0.21	0.01	-0.35	142.46	-4.91
drink-juice	-0.42	0	-13.81	-0.33	-0.09	-172.86	-0.03	-0.03	0	-0.08	-0.24	-0.68
drink-coffee	-0.39	0	-3.09	-0.03	-0.16	144.09	-0.01	-1.34	0	-0.02		-2.15
dressings	0.09	0	-5.68	0.13	0.01	-66.14	-0.02	-0.13	0	0.03	96.01	2.42
dairy-yogurt-drink	-0.01	0	-0.41	0	0	-0.89	0	-0.02	0	0	21.01	-0.71
culinary-ingredients	-1.35	0	2.69	0.02	0.01	102.79	-0.05	-0.01	-0.01	0.01	2.68	-24.14
cookies-biscuit	-0.22	0	-0.28	-0.01	0	-8.41	-0.01	-0.28	0	0	-4.26	-8.76
coffee-beans-wf	-0.68	-0.06		30.12		-18.32	3.16	-76.14		38.27		
cheese	-0.09	0	-0.06	0	-0.16	1160.41	0	0	0	0	120.24	10.91
cereal	0.16	0	1.99	0	0	-10.46	0	-0.01	0	0	46.46	-20.86
canned-goods	-0.51	0	19.99	0.01	0	-35.89	0	0	0	0		106.37
cakes	-0.32	0	-0.19	0	0	-21.5	0	0	0	0	88.6	-21.04
breakfast	-0.22	0	0.31	0	0.01	-3.93	0	0.01	0.01	0	-26.05	1.56
bread	-3.84	0	0.21	0	0	-23.24	-0.01	0	0	0	-9.95	-15.14
baking	-0.1	0	-0.02	0	0	-16.16	0	0	0	0	299.38	-3.77
baby-food	4.86	0.02	69.47	0.4	0.03	30.27	0.06	4.87	0.03	0.14	-55.89	-3.59

Nutrient

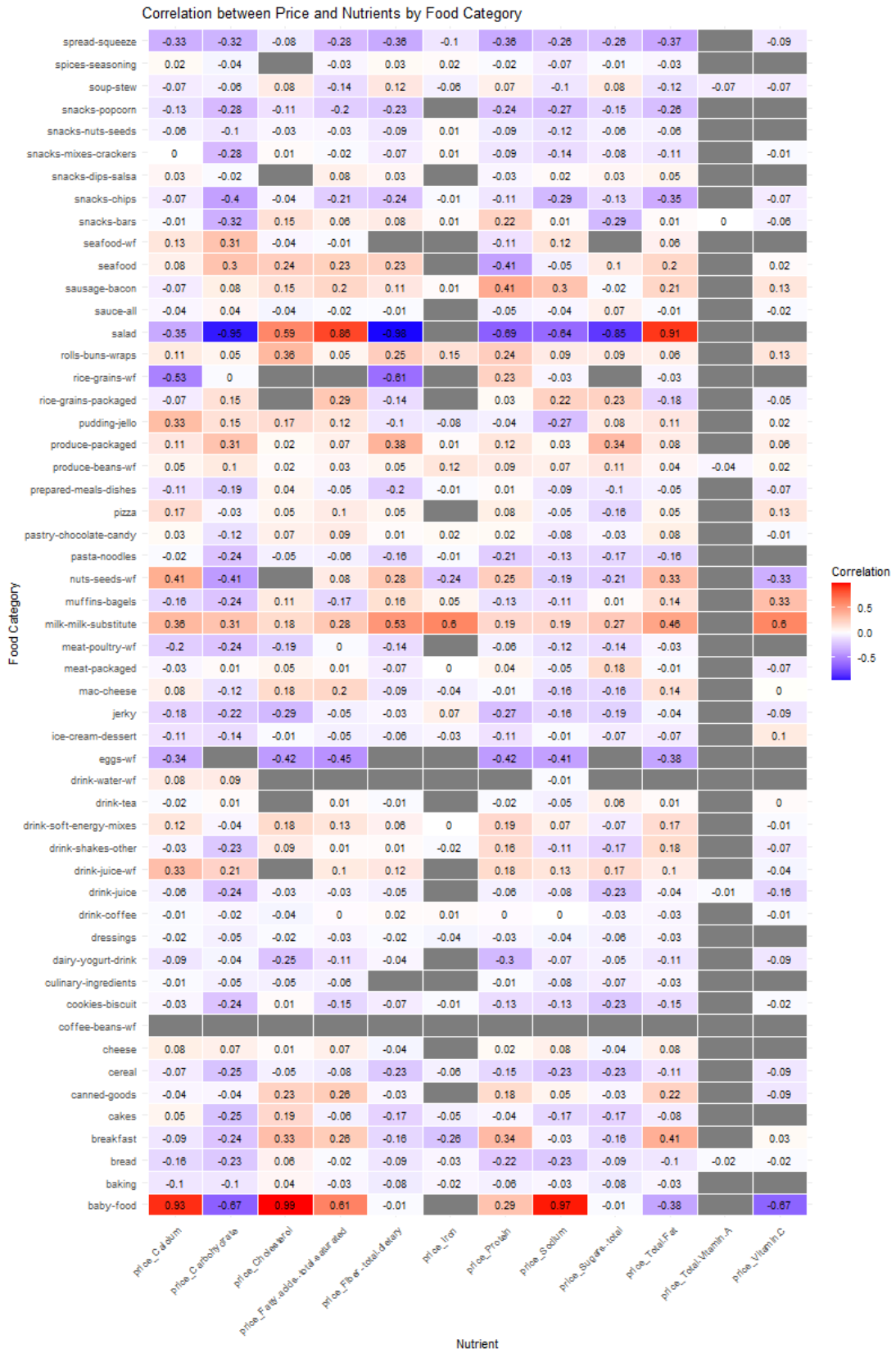
Slope

800

400

0

Heatmaps for Correlation and Regression for Target



Regression Slopes of Price on Nutrients by Food Category

Food Category	Calcium	Carbohydrate	Cholesterol	Fatty acids, total saturated	Fiber, total dietary	Iron	Protein	Sodium	Sugars, total	Total Fat	Total Vitamin A	Vitamin C
spread-squeeze	-14.42	-0.05	-82.64	-0.19	-0.39	-261.93	-0.14	-5.2	-0.07	-0.07		-71.04
spices-seasoning	7.02	-0.01		-0.1	0.05	45.81	-0.05	-0.08	-0.03	-0.06		
soup-stew	-0.18	0	16.08	-0.07	0.11	-169.39	0.02	-0.06	0.03	-0.03	-0.19	-0.19
snacks-popcorn	-40.79	-0.14	-254.17	-0.69	-0.68		-0.82	-9.82	-0.24	-0.23		
snacks-nuts-seeds	-4.8	-0.05	-87.72	-0.06	-0.19	23.43	-0.08	-4.02	-0.06	-0.02		
snacks-mixes-crackers	-0.02	-0.04	1.04	-0.01	-0.07	5.8	-0.04	-1.03	-0.02	-0.03		-0.74
snacks-dips-salsa	4.1	0		0.17	0.04		-0.02	0.22	0.02	0.02		
snacks-chips	-6.66	-0.14	-18.78	-0.39	-0.62	-2.57	-0.09	-6.18	-0.24	-0.21		-41.94
snacks-bars	-0.78	-0.11	86.21	0.1	0.09	11.25	0.15	0	-0.14	0.01	3.44	-30.58
seafood-wf	18.07	1.82	-6.45	-0.06			-0.12	4.7		0.05		
seafood	5.05	0.17	22.07	0.81	1.57		-0.23	-0.46	0.18	0.16		29.2
sausage-bacon	-2.01	0.04	9.07	0.08	0.42	13.81	0.09	1.15	-0.02	0.03		32.64
sauce-all	-5.97	0.02	-32.81	-0.05	-0.03		-0.08	-0.18	0.04	0		-9.5
salad	-11.06	-0.05	13.4	0.68	-0.7		-0.67	-7.63	-0.97	0.26		
rolls-buns-wraps	0.88	0	36.58	0.02	0.04	141.75	0.05	0.34	0.02	0.01		8.15
rice-grains-wf	-54.17	0			-0.91		0.19	-5		-0.21		
rice-grains-packaged	-1.18	0		1.09	-0.03		0.01	0.45	0.23	-0.03		-18.69
pudding-jello	10.34	0.01	69.53	0.1	-0.08	-58	0	-0.45	0	0.05		19.51
produce-packaged	10.2	0.06	11.01	0.14	0.57	78.91	0.18	0.17	0.1	0.05		8.49
produce-beans-wf	4.79	0.01	2.56	0.47	0.04	343.58	0.15	5.07	0.02	0.16	-2.84	6.13
prepared-meals-dishes	-4.08	-0.04	3.86	-0.06	-0.43	-59.61	0.01	-0.99	-0.07	-0.02		-53.47
pizza	4.51	-0.01	7.91	0.12	0.08		0.05	-0.6	-0.12	0.03		121.45
pastry-chocolate-candy	1.09	-0.01	22.55	0.03	0	29.51	0.02	-1.71	0	0.02		-21.27
pasta-noodles	-0.05	-0.02	-48.37	-0.37	-0.19	-14.23	-0.1	-2.22	-0.33	-0.33		
nuts-seeds-wf	21.19	-0.35		0.17	0.49	-325	0.29	-3.88	-0.59	0.14		-790.67
muffins-bagels	-2.59	-0.02	7.16	-0.11	0.07	49.89	-0.04	-0.93	0	0.03		305.27
milk-milk-substitute	7.18	0.14	120.44	0.99	5.62	2467.76	0.18	10.84	0.14	0.75		493.55
meat-poultry-wf	-47.3	-1.77	-15.35	-0.01	-1.88		-0.03	-2.24	-9.4	-0.02		
meat-packaged	-2.39	0	7.49	0.01	-0.22	-6.38	0.02	-0.49	0.39	0		-86.95
mac-cheese	0.6	-0.01	18.36	0.15	-0.09	-58.37	0	-0.94	-0.13	0.05		2.86
jerky	-11.93	-0.15	-58.4	-0.07	-0.35	374.43	-0.14	-1.77	-0.13	-0.02		-59.31
ice-cream-dessert	-2.77	-0.01	0	-0.02	-0.07	-8.53	-0.08	0	-0.01	-0.01		15.4
eggs-wf	-24.83		-4.91	-0.63			-0.15	-12.72		-0.17		
drink-water-wf	72.4	10.25						-13.09				
drink-tea	-4.78	0		4.67	-0.01		-0.06	-3.99	0.02	0.67		0.34
drink-soft-energy-mixes	2.34	-0.02	431.36	4.16	0.96	-5.65	0.36	0.53	-0.05	2.88		-0.52
drink-shakes-other	-1.25	-0.08	61.65	0.05	0.08	-95.18	0.28	-4.07	-0.08	0.77		-2.04
drink-juice-wf	67.02	0.05		1.62	1.09		0.93	22.5	0.05	1.16		-4.75
drink-juice	-55.16	-1.22	-919.83	-2.53	-6.1		-2.79	-75.88	-1.25	-4.46	-101.51	-331.84
drink-coffee	-0.42	-0.02	-385.54	-0.01	0.25	282.73	-0.01	0.04	-0.02	-0.07		-0.76
dressings	-0.4	-0.01	-3.56	-0.02	-0.12	-149.6	-0.01	-0.07	-0.02	-0.01		
dairy-yogurt-drink	-1.87	-0.01	-57.84	-0.09	-0.11		-0.13	-3.72	-0.01	-0.07		-15.03
culinary-ingredients	-1.45	-0.01	-12.25	-0.05			-0.19	-0.13	-0.01	-0.01		
cookies-biscuit	-1.56	-0.04	1.02	-0.1	-0.12	-27.75	-0.18	-2.89	-0.06	-0.06		-14.08
coffee-beans-wf												
cheese	0.35	0.03	0.31	0.01	-0.03		0	0.35	-0.03	0.01		
cereal	-1.2	-0.01	-34.78	-0.1	-0.03	-16.81	-0.04	-1.91	-0.03	-0.04		-50.22
canned-goods	-8.7	-0.01	31.15	0.86	-0.01		0.09	0.23	-0.01	0.17		-126.53
cakes	2.94	-0.03	28.68	-0.05	-0.03	-248.12	-0.07	-3.56	-0.03	-0.03		
breakfast	-0.99	-0.02	7.9	0.2	-0.04	-105.64	0.1	-0.28	-0.04	0.1		7.77
bread	-6.79	-0.06	37.34	-0.06	-0.08	-125.48	-0.26	-5.53	-0.08	-0.14	-10.27	-84.14
baking	-6.78	-0.01	29.17	-0.02	-0.02	-35.44	-0.07	-0.07	-0.02	-0.01		
baby-food	10.67	-0.68	255	0.95	-0.01		0.22	36.08	-0.01	-0.19		-105

Slope

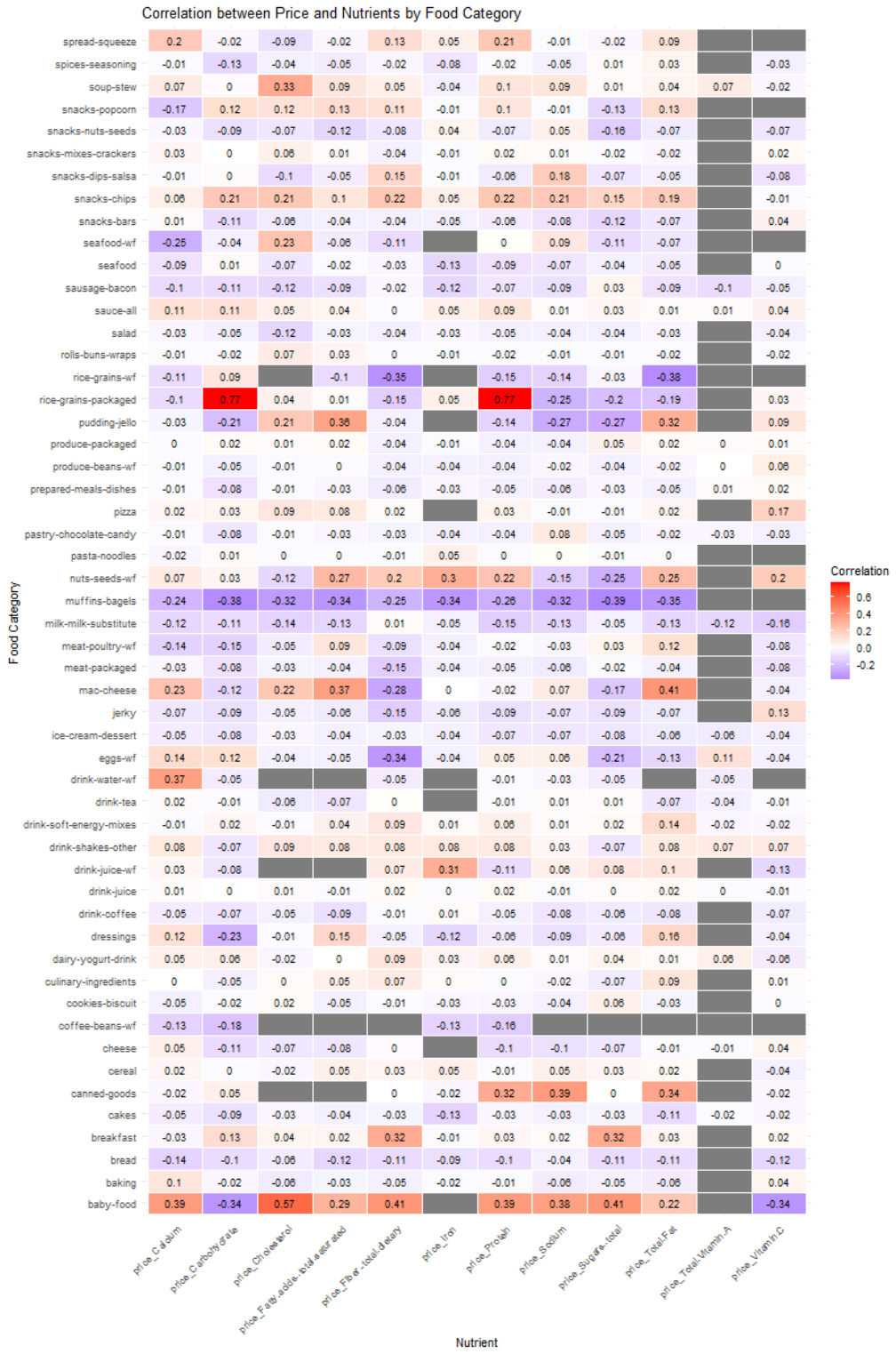
2000

1000

0

Nutrient

Heatmaps for Correlation and Regression for Walmart



Regression Slopes of Price on Nutrients by Food Category

Food Category	Calcium	Carbohydrate	Cholesterol	Fatty acids--total saturated	Fiber--total dietary	Iron	Protein	Sodium	Sugars--total	Total Fat	Total Vitamin A	Vitamin C
spread-squeeze	8.53	0	-8.02	-0.01	0.09	105.08	0.06	-0.13	0	0.01		
spices-seasoning	-0.22	0	-2.18	0	0	-39.73	0	0	0	0.01		-4.4
soup-stew	0.17	0	0.77	0.01	0	-1.63	0	0.01	0	0	46.13	-1.68
snacks-popcorn	-4.01	0	0.02	0	0.01	-2.85	0	0	0	0		
snacks-nuts-seeds	-0.08	0	-42.76	-0.01	0	8.4	0	0	0	0		-24.63
snacks-mixes-crackers	0.23	0	4.22	0	-0.01	-1.18	0	0.02	0	0		0.81
snacks-dips-salsa	-0.03	0	-1.83	0	0.03	-16.44	0	0.12	0	0		-8.07
snacks-chips	0.5	0	0.21	0	0.02	14.84	0.01	0	0.01	0		-0.16
snacks-bars	0.03	0	-2.64	0	0	-13.47	0	-0.37	0	0		14.74
seafood-wf	-17.41	-0.05	50.04	-0.29	-0.57		0	0.02	-0.29	-0.1		
seafood	-0.42	0	-0.11	0	-0.01	-36.78	0	-0.01	-0.01	0		-2.07
sausage-bacon	-0.58	-0.01	-0.74	0	-0.07	-45.51	0	-0.05	0	0	-331.4	-1.65
sauce-all	0.15	0	0.07	0	0	11.95	0	0	0	0	37.94	4.49
salad	-0.07	0	-8.95	0	0	-2.75	0	-0.01	0	0		-2.03
rolls-buns-wraps	-0.01	0	5.17	0	0	-0.6	0	-0.01	0	0		-0.33
rice-grains-wf	-16.07	0.12		-12.88	-0.88		-0.54	-143.22	-0.35	-1.68		
rice-grains-packaged	-10.8	0	89.66	0.02	-0.29	56.04	0.01	-1.84	-0.52	-0.25		6.47
pudding-jello	-0.41	-0.01	11.07	0.15	-0.1		-0.01	-0.56	-0.01	0.09		49.78
produce-packaged	-0.01	0	0.53	0	0	-3.72	0	0	0	0	4.65	0.05
produce-beans-wf	-0.1	0	-6.39	-0.01	0	-18.72	0	0	0	-0.01	-0.69	2.39
prepared-meals-dishes	-0.01	0	-0.05	0	0	-4.67	0	0	0	0	16.29	0.29
pizza	0.49	0	0.75	0	0		0	0	0	0		81.64
pastry-chocolate-candy	-0.03	0	-0.06	0	0	-7.19	0	0.01	0	0	-225.4	-2.46
pasta-noodles	-0.05	0	0	0	0	1.9	0	0	0	0		
nuts-seeds-wf	1.27	0	-15.55	0.26	0.07	269.86	0.07	-0.18	-0.03	0.03		48
muffins-bagels	-0.08	0	-0.59	0	-0.01	-11.09	0	-0.04	0	0		
milk-milk-substitute	-0.02	0	-0.45	0	0	-10.67	0	-0.05	0	0	-62.21	-5.8
meat-poultry-wf	-68.79	-0.13	-4.23	0.09	-0.44	-36.37	0	-0.58	0.24	0.07		-131.01
meat-packaged	-0.24	-0.02	-0.6	-0.01	-0.43	-79.57	0	-0.07	-0.01	0		-94.96
mac-cheese	3.93	-0.01	40.66	0.44	-0.48	1.58	-0.01	0.46	-0.07	0.2		-19.05
jerky	-0.88	0	-0.03	0	-0.32	-5	0	-0.01	0	0		32.35
ice-cream-dessert	-0.05	0	0	0	0	-12.76	0	-0.09	0	0	-102.04	-0.69
eggs-wf	0.03	0	-0.02	0	-1.47	-3.83	0	0.07	-0.07	0	55.83	-40.09
drink-water-wf	12.57	0			0		-0.96	-0.5	-0.01		-11.18	
drink-tea	0.79	0	-26.87	-0.03	0.01		-0.02	0.08	0	-0.03	-92.55	-0.26
drink-soft-energy-mixes	-0.04	0	-1.1	0	0.05	34.12	0.02	0	0	0.03	-124.47	-0.19
drink-shakes-other	0.1	0	6.98	0.03	0.01	18.16	0	0.07	0	0.01	117.45	0.7
drink-juice-wf	0.21	0			0.02	338.82	-0.03	0.67	0.01	1.02		-1.62
drink-juice	0.06	0	3.42	-0.03	0.02	-12.14	0.01	-0.01	0	0.02	-10.76	-0.04
drink-coffee	-0.22	0	-1.88	-0.02	-0.03	42.45	0	-0.85	0	-0.01		-13.39
dressings	0.75	-0.02	-0.41	0.03	-0.06	-209.24	-0.01	-0.08	0	0.01		-8.69
dairy-yogurt-drink	0.01	0	-0.09	0	0.01	5.67	0	0.01	0	0	42.13	-4.3
culinary-ingredients	-0.02	0	0.1	0	0.01	6.32	0	0	0	0		10.15
cookies-biscuit	-0.14	0	0.24	0	0	-2.8	0	-0.05	0	0		-1.74
coffee-beans-wf	-0.53	-0.05				-14.33	-12.16					
cheese	0.04	0	-0.03	0	-0.02		0	0	0	0	-17.13	55.33
cereal	0.13	0	-0.87	0	0	3.29	0	0.02	0	0		-3.6
canned-goods	-0.04	0			0	-1.24	0	0	0	0		-2.07
cakes	-0.23	0	-0.12	0	0	-17.23	0	0	0	0	-120.81	-12.4
breakfast	-0.16	0	0.2	0	0.01	-0.62	0	0.01	0.01	0		2.94
bread	-0.92	0	-0.69	0	0	-10.96	0	0	0	0		-9.23
baking	0.92	0	-0.01	0	0	-4.7	0	0	0	0		11.8
baby-food	5.38	-0.22	49.09	0.35	0.09		0.29	12.72	0.09	0.22		-13.57

Nutrient

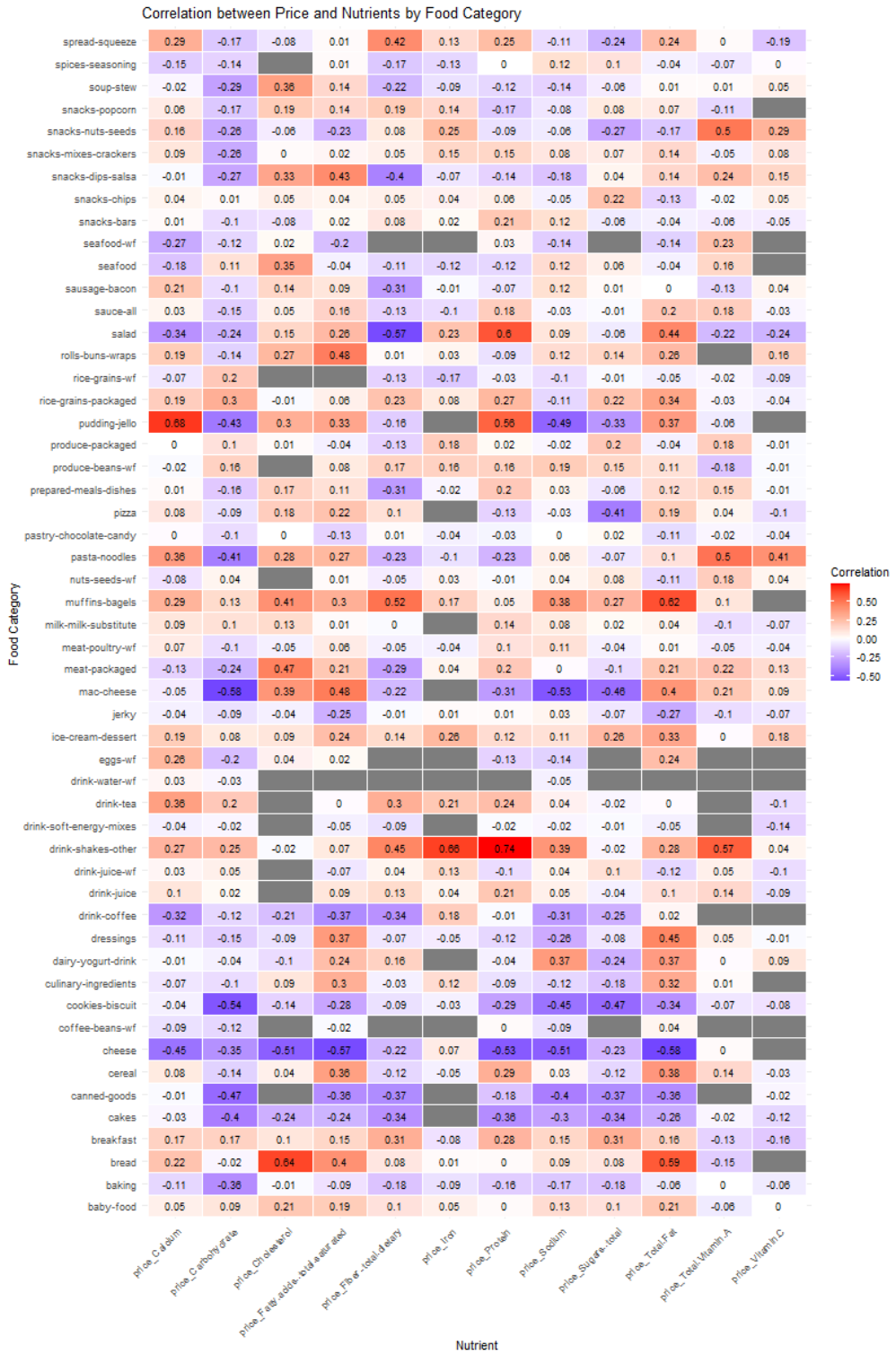
Slope

200

0

-200

Heatmaps for Correlation and Regression for Whole Foods



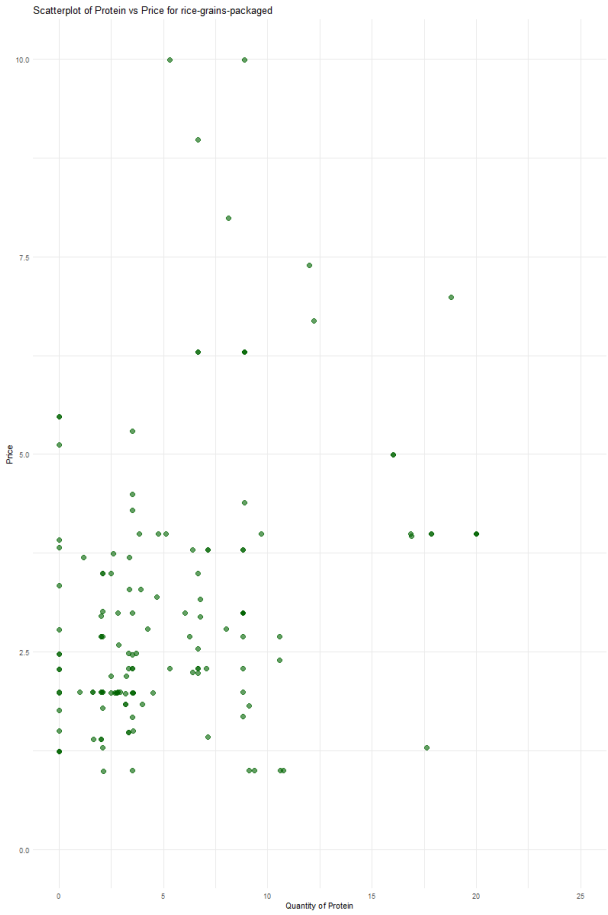
Regression Slopes of Price on Nutrients by Food Category

Food Category	Calcium	Carbohydrate	Cholesterol	Fatty.acids..total.saturated	Fiber..total.dietary	Iron	Protein	Sodium	Sugars..total	Total.Fat	Total.Vitamin.A	Vitamin.C
spread-squeeze	8.21	-0.04	-24.45	0.01	0.42	211.83	0.1	-2.85	-0.05	0.04	7.19	-35.48
spices-seasoning	-1.22	-0.01		0.02	-0.05	-22.18	0	0.06	0.02	-0.02	-1.71	-0.1
soup-stew	-1.38	-0.03	73.08	0.15	-0.16	-96.65	-0.03	-0.22	-0.03	0	4.75	197.01
snacks-popcorn	1.42	-0.02	19.61	0.03	0.03	87.02	-0.04	-0.3	0.01	0.01	-97.59	
snacks-nuts-seeds	8.72	-0.08	-4.49	-0.42	0.07	255.46	-0.08	-1.93	-0.1	-0.06	572.22	67.2
snacks-mixes-crackers	0.65	-0.02	0.21	0.01	0.01	62.88	0.03	0.34	0.01	0.02	-3.82	6.32
snacks-dips-salsa	-0.26	-0.04	23.12	0.32	-0.21	-68.01	-0.06	-0.48	0.01	0.02	19.17	16.18
snacks-chips	0.44	0	3.69	0.01	0.02	14.85	0.01	-0.16	0.04	-0.03	-5.75	6.2
snacks-bars	0.9	-0.07	-40.82	0.04	0.12	39.81	0.22	7.06	-0.05	-0.04	-24.88	-33.73
seafood-wf	-46.66	-0.78	3.34	-1.64			0.06	-3.93		-0.28	137.97	
seafood	-6.21	0.1	39.47	-0.12	-0.87	-266.06	-0.06	0.51	0.19	-0.03	293.54	
sausage-bacon	6.37	-0.03	6.39	0.03	-0.32	-19.16	-0.02	0.4	0.01	0	-80.48	12.45
sauce-all	1.12	-0.03	13.08	0.18	-0.25	-366.32	0.17	-0.05	0	0.06	85.64	-6.88
salad	-22.21	-0.09	5.25	0.44	-0.55	380.51	0.3	1.17	-0.05	0.17	-57.65	-15.86
rolls-buns-wraps	3.28	-0.02	24.77	0.54	0	21.46	-0.04	0.34	0.05	0.07		57.05
rice-grains-wf	-5.27	0.05			-0.07	-155.55	-0.02	-47.46	-0.02	-0.09	-65.76	-51.73
rice-grains-packaged	11.59	0.02	-8.82	0.28	0.14	62.56	0.09	-0.48	0.22	0.39	-38	-13.63
pudding-jello	23.27	-0.03	10.07	0.11	-0.25		0.6	-4.96	-0.03	0.08	-14.95	
produce-packaged	0.02	0.02	8.22	-0.03	-0.17	77.54	0.01	-0.1	0.05	-0.02	9.95	-0.58
produce-beans-wf	-0.5	0.02		0.08	0.07	82.21	0.06	1.5	0.04	0.07	-4.07	-0.22
prepared-meals-dishes	0.69	-0.04	15.21	0.12	-0.63	-49.99	0.15	0.74	-0.07	0.07	105	-5.04
pizza	3.34	-0.04	17.16	0.23	0.29		-0.15	-0.6	-0.93	0.09	32.05	-76.24
pastry-chocolate-candy	-0.07	-0.01	-0.06	-0.04	0	-23.16	-0.02	-0.03	0	-0.02	-34.49	-1.28
pasta-noodles	17.04	-0.04	28.75	0.92	-0.14	-77.76	-0.08	0.41	-0.08	0.13	570.72	282.08
nuts-seeds-wf	-4.07	0.01		0.02	-0.03	11.89	0	1.07	0.02	-0.04	7.59	209.56
muffins-bagels	5.73	0.01	43.97	0.25	0.13	132.55	0.02	2.86	0.04	0.16	54.81	
milk-milk-substitute	1.66	0.03	16.38	0.01	-0.01		0.09	3.83	0.01	0.01	-36.13	-204.19
meat-poultry-wf	17.72	-0.24	-6.3	0.11	-1.27	-241.04	0.08	1.2	-0.67	0.01	-5.51	-241.04
meat-packaged	-4.29	-0.07	35.36	0.15	-0.44	78.39	0.04	0	-0.09	0.06	79.73	57.74
mac-cheese	-2.04	-0.07	100.81	0.73	-0.2		-0.14	-5.17	-0.37	0.32	174.98	32.54
jerky	-2.46	-0.03	-4.56	-0.23	-0.01	17.22	0	0.26	-0.03	-0.12	-110.66	-205.77
ice-cream-dessert	5.06	0.01	4.35	0.09	0.12	762.02	0.08	1.96	0.03	0.07	-0.99	108.49
eggs-wf	26.8	-0.47	0.93	0.04			-0.39	-53.02		0.2		
drink-water-wf	22.36	-3.51						-205.63				
drink-tea	41.42	0.1		-0.01	0.38	681.12	0.57	5.06	-0.02	0		-47.31
drink-soft-energy-mixes	-45.54	-0.02		-4.57	-0.78		-0.3	-18.32	-0.01	-4.57		-147.52
drink-shakes-other	16.5	0.14	-96.17	0.24	0.63	2018.08	0.5	12.78	-0.02	0.6	820.18	11.68
drink-juice-wf	11.63	0.02		-0.57	0.47	95.31	-0.71	8.16	0.07	-0.49	86.91	-20.88
drink-juice	6.53	0.01		0.46	0.62	54.11	0.82	4.17	-0.02	0.37	20.04	-7.4
drink-coffee	-23.44	-0.05	-96.65	-0.94	-3.93	303.17	-0.03	-24.95	-0.18	0.02		
dressings	-8.55	-0.04	-21.92	0.23	-0.14	-165.93	-0.12	-0.95	-0.03	0.06	38.9	-7.1
dairy-yogurt-drink	-0.15	-0.01	-24.86	0.11	0.28		-0.02	6.04	-0.07	0.1	-1.61	6.75
culinary-ingredients	-9.69	-0.01	4.63	0.05	-0.02	151.7	-0.15	-0.04	-0.03	0.03	0.29	
cookies-biscuit	-1.58	-0.05	-10.5	-0.09	-0.05	-31.51	-0.16	-5.35	-0.07	-0.07	-21.4	-26.21
coffee-beans-wf	-123.46	-4.77		-12.87			0.13	-123.46		5.99		
cheese	-8.08	-0.39	-72.92	-0.43	-0.76	811.5	-0.3	-8.42	-0.63	-0.27	1.85	
cereal	1.5	-0.01	3.58	0.21	-0.03	-15.75	0.16	0.29	-0.03	0.08	36.41	-14.36
canned-goods	-1.25	-0.1		-0.25	-0.16		-0.1	-1.63	-0.16	-0.17		-6.67
cakes	-3.97	-0.1	-24.99	-0.23	-0.12		-0.9	-11.36	-0.12	-0.15	-15.77	-713.36
breakfast	1.61	0.01	2.02	0.06	0.03	-78.36	0.06	0.79	0.03	0.03	-44.92	-45.62
bread	9.82	0	55.6	0.56	0.03	19.5	0	0.92	0.03	0.21	-125.67	
baking	-0.91	-0.06	-7.54	-0.06	-0.04	-130.75	-0.1	-0.26	-0.04	-0.02	-5.3	-11.16
baby-food	1.85	0.02	78.43	0.36	0.03	40.65	0	3.57	0.03	0.14	-51.04	0.01

Slope: 2000, 1000, 0

Nutrient

Sample Scatterplots



Insights

From the correlation heatmaps, I gained a better idea of how each nutrient might affect the pricing of certain types of products. Although, most of the scores indicate that nutrients may have very little impact on pricing. There are some standouts, however, that have the potential to be looked into as there is some positive correlation between them. The correlation isn't too strong as they're closer to 0.5-0.6 rather than being more concrete though.

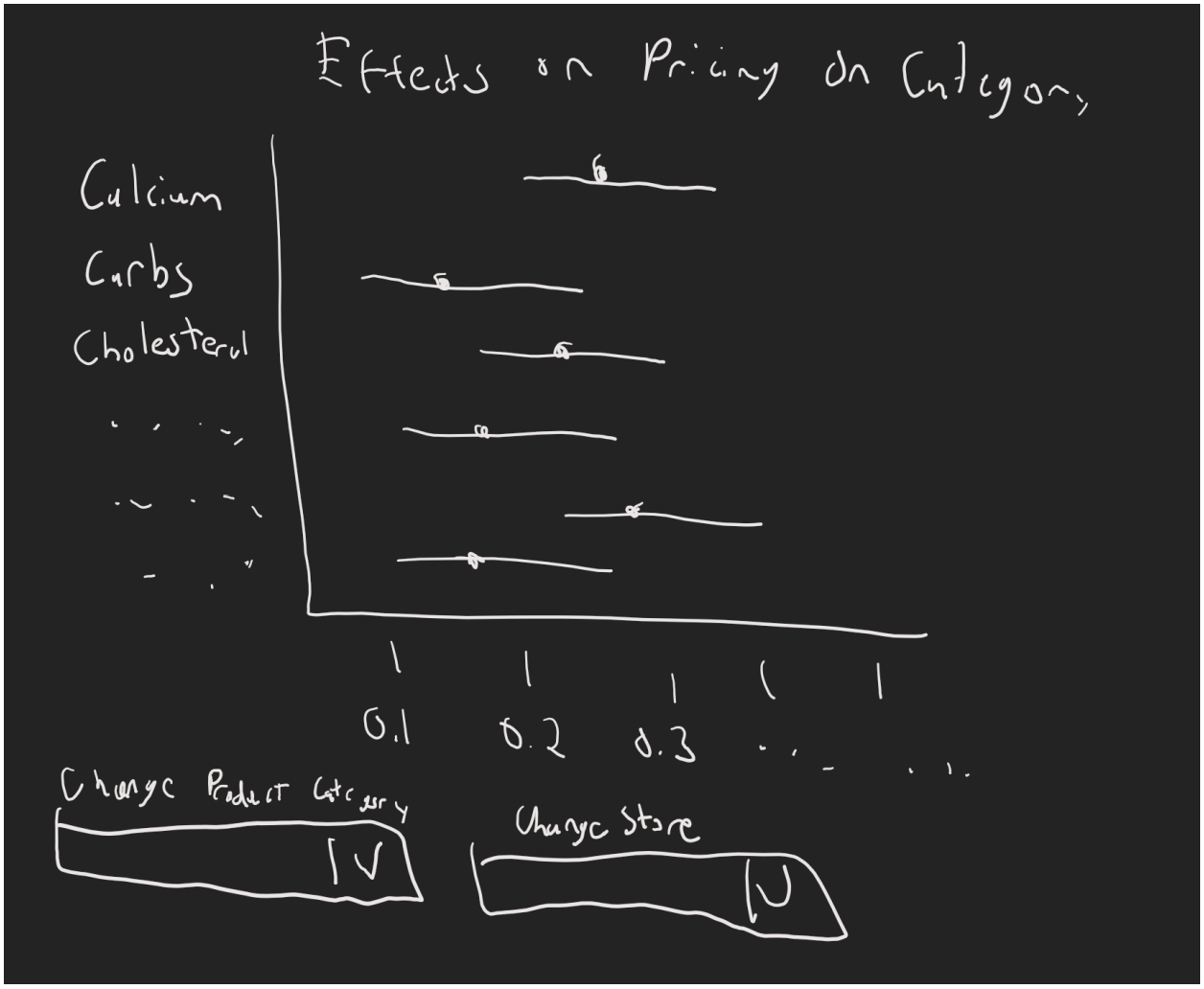
From the regression heatmaps, I estimated a general overlook on how nutrients could affect pricing, if there was an actual impact on them to help get a better idea on what some potential targets that I could look at. I also gained a better idea of what relationships I might want to disregard. For instance, there are a bunch of regressions that have extremely high price changes (ex: 1000) which is influenced by there being columns of data (in nutrients) that are mostly 0s, so fluctuations cause larger impacts in them. This information doesn't help much as there isn't much data or variety to work with there, and overall just feels unreliable to make assumptions on.

From the scatterplots, I had some generated with more simplicity before the actual implementation to see how they would look. I chose some with stronger correlations from the heatmaps as a test. They do look like that there could be a potential increase in pricing with an increase in a particular nutrient, with outliers here and there.

After all of this I probably can trim stuff for the next design as I should probably cut or limit features. Some nutrients have very little or unreliable impact to the question in hand (ex: mostly 0s) so I am going to be removing those as options on the final visualization. The dataset is also large, it may be difficult to render or even see all the data points as they enumerate in the ten thousands. Which enforces the need to be able to have the user split the data into subsets that they might want to view at a time. So definitely I would need the ability for the user to turn on/off the ability to see certain stores.

Design Evolution

This is the initial conceptualization for the scatterplot. The user has the ability to switch between the types of nutrients and categories. They can also see the distribution of items related to stores, based on colors. The user will be given the option to enable or disable what items from what stores are displayed. Finally, you can hover over a dot to see more information about that specific point. It's mostly a simple scatterplot to cover as many bases as possible.



This is the initial conceptualization for the coefficient plot to help show the impact of each nutrient on pricing. The user can see each of the different nutrients on the sides and see the overall coefficients on the the bottoms. As there are a lot of different factors at play that would require multiple plots, the user can switch between them by changing product categories and stores so hundreds of different plots don't need to be rendered all at once.

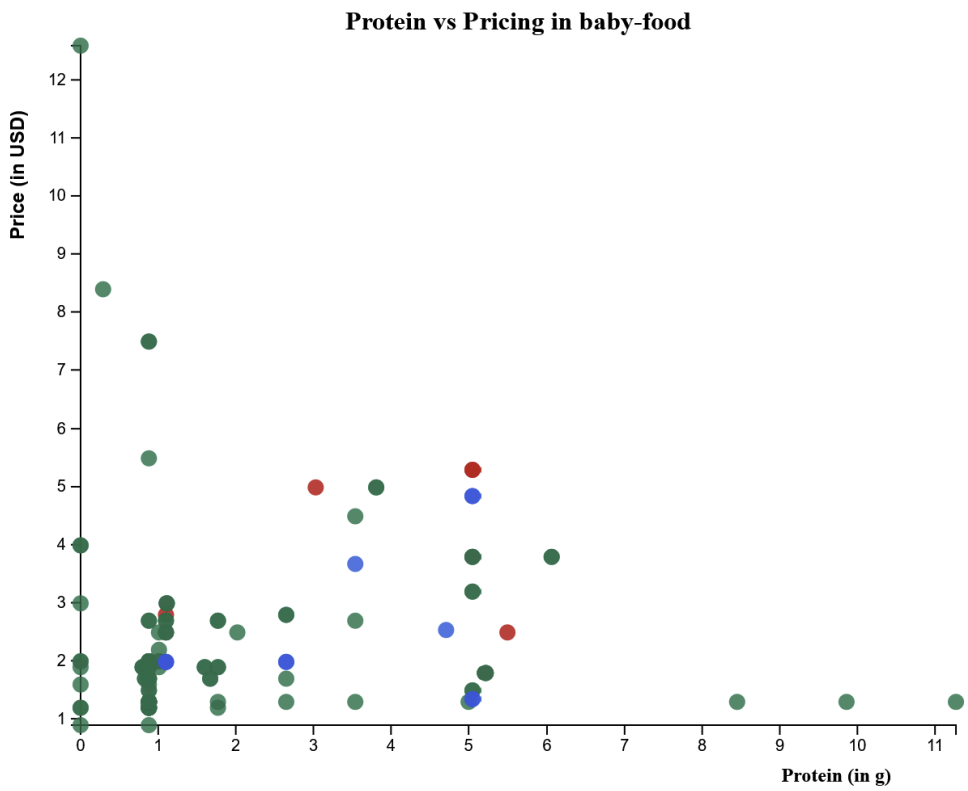
The design overall did not deviate much from the proposal. This is due to me not finding anything drastic that would require me to completely change how I would need to visualize the data. I also did not find any particularly exciting patterns that I wanted to capture with a new type of visual that couldn't be captured with the existing visuals. Additionally, I am the sole student working on this project and had limited time due to a busy MQP, so I could not increase the scope of this project.

In the end, I felt like some of the design decisions I made were simple enough, but provided a strong experience to quickly and easily provide visual information to the user.

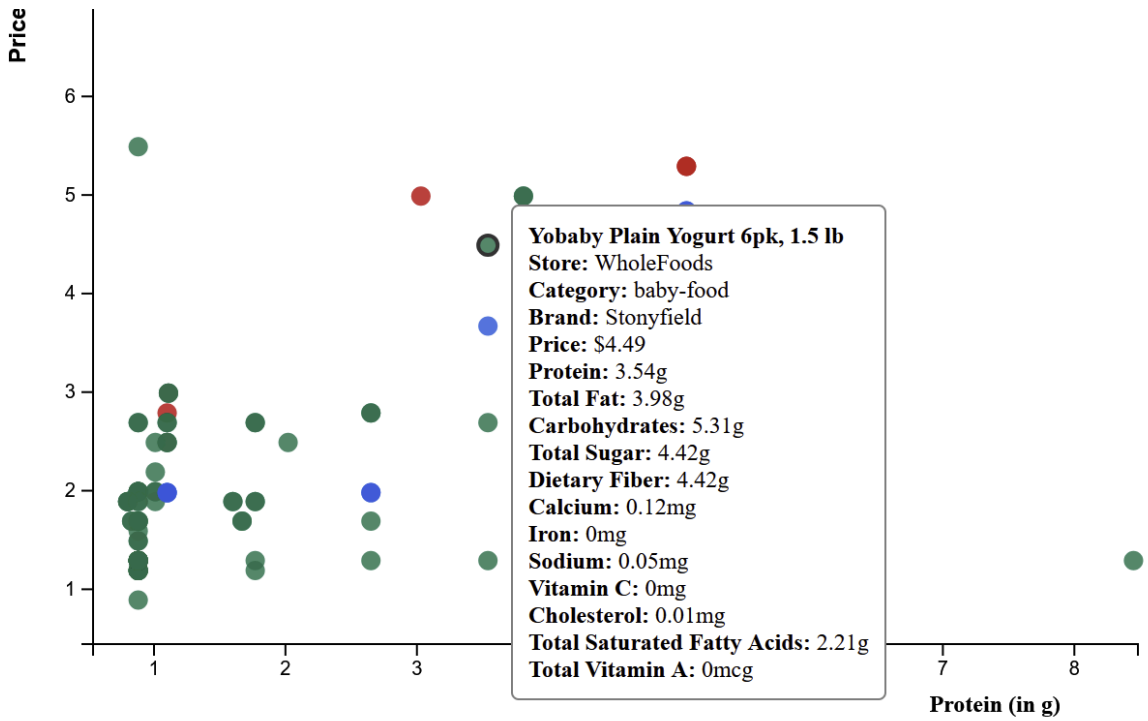
One of which was the color choice for the points on the scatterplots. Red, blue, and green are distinct enough colors from each other that it's easy to tell that they are different from a glance. Additionally, each color matches the logo of the respective store brand, making it easy to associate it with their respective classification.

Another of was the usage of a coefficient plot. I felt like a coefficient plot had some good utility to easily convey scale and differences of the impacts of nutrients. Utilizing length as a medium is simplistic and something that humans can quickly understand, which helps to convey the information from this graphic. I could additionally use markers at the ends of each line to better convey the start and stop points. The horizontal parallel lines helped viewers to compare values vertically across categories easily due to their positioning.

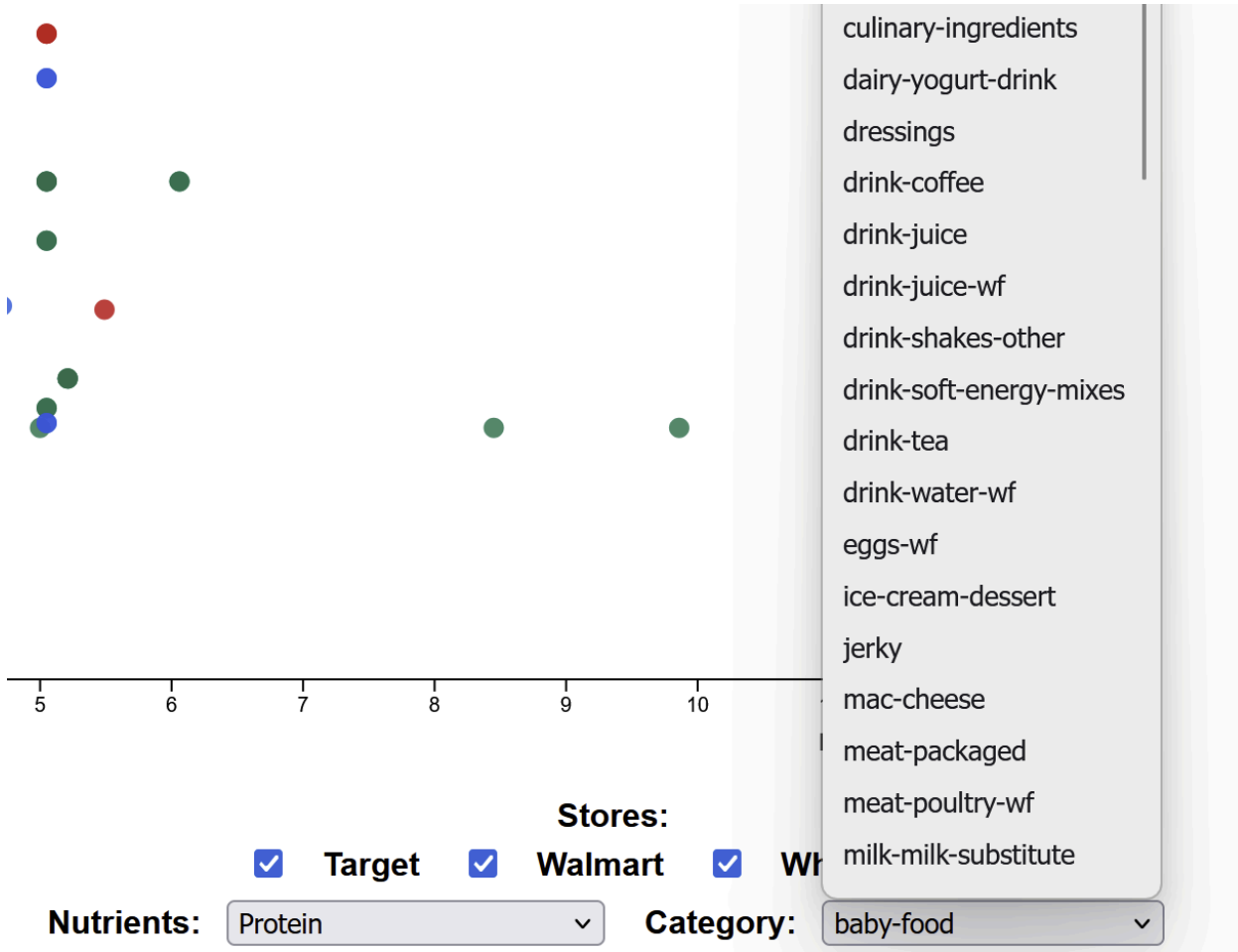
Implementation



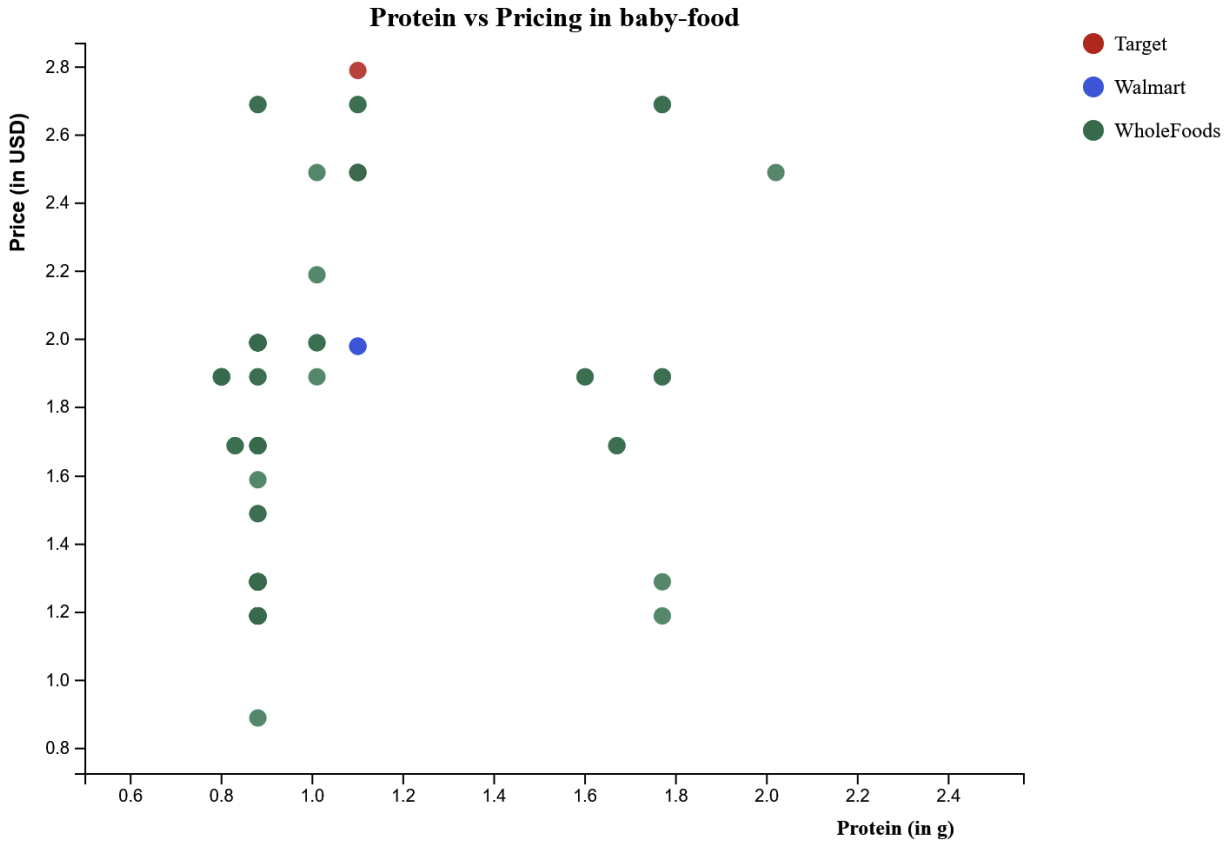
The first visualization is a scatterplot that aims to show the distribution of items to compare the pricing of the product to nutrients to gleam any potential patterns. Each individual dot represents one item and they are color coded based on the store listing. The colorings are red for Target, blue for Walmart, and green for Whole Foods.



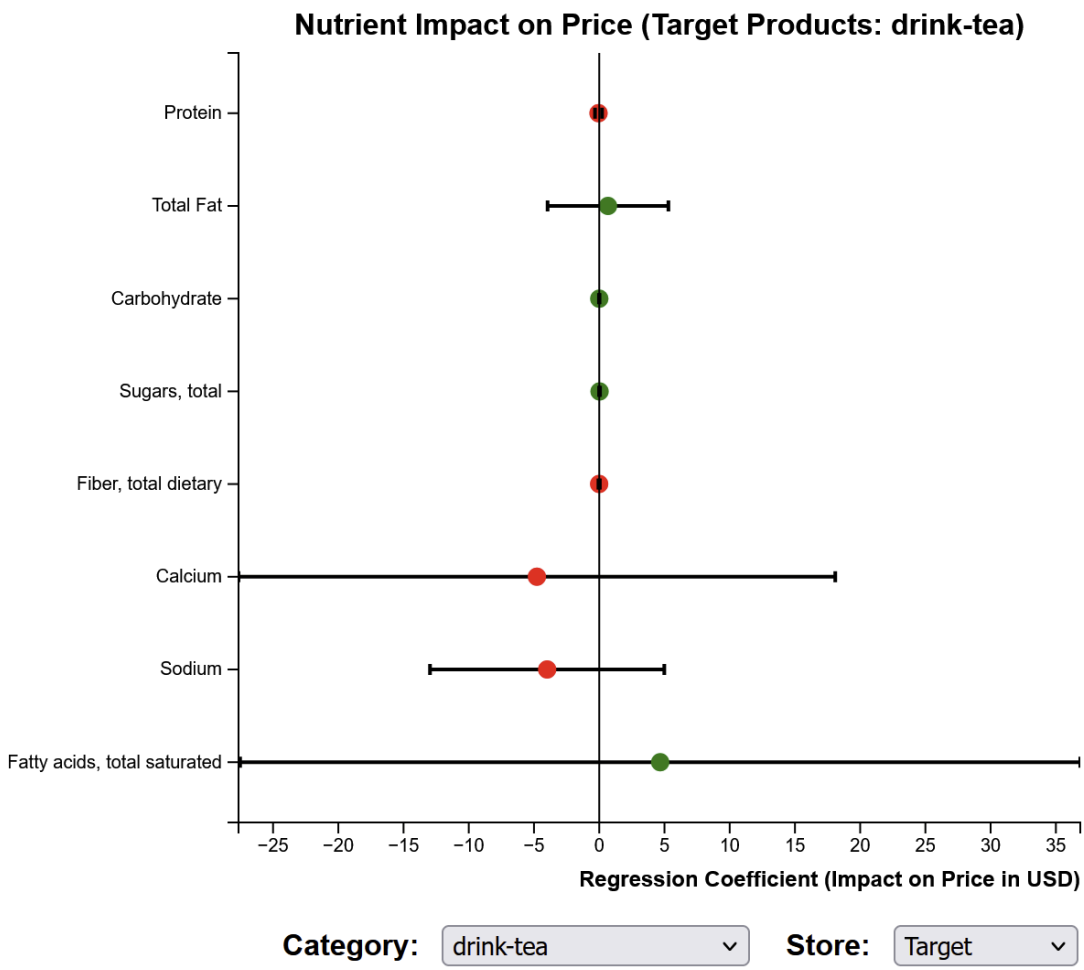
Hovering over any point opens a tooltip giving the exact information about the item. This is to help users identify what a product might be in a specific location on the scatterplot, to grant the exact values for the price and nutrients, and maybe also give information about other nutrients of the product the user may be interested in.



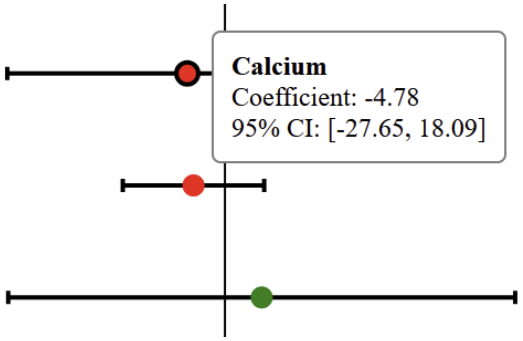
The drop down boxes allow the user to alter what data is used on the scatter plots. The user can select what kind of category they want to compare nutrient values to pricings. There is also the ability to enable and disable what stores are shown. This is partially to help the user choose whether they want to reduce the lag from creating and managing all the data points when interacting with the scatterplot. Additionally, it helps them better visualize any patterns just for one particular store, if they so desire.



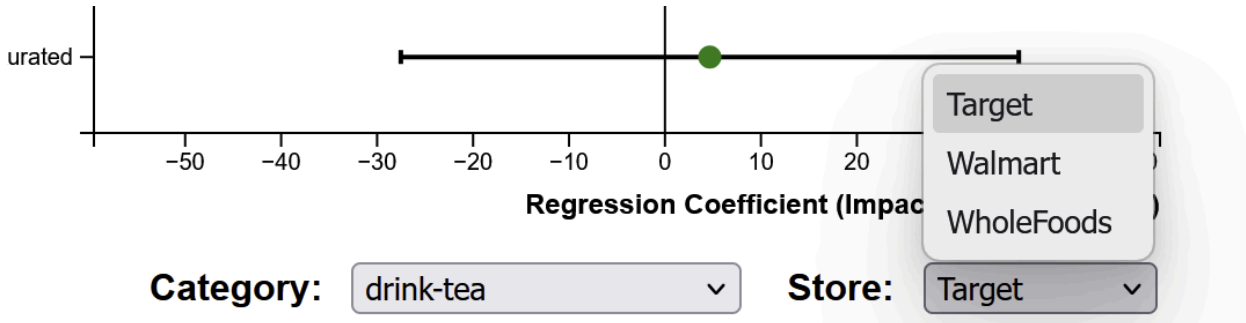
The coefficient plot also has a zoom and panning feature, when you scroll on the visual or drag it around. This was implemented to help users see and interact with the points more easily. Additionally, many of the points get clustered together (due to the high number of data points) making it difficult to see from a distance, so the zoom feature helps to solve that.



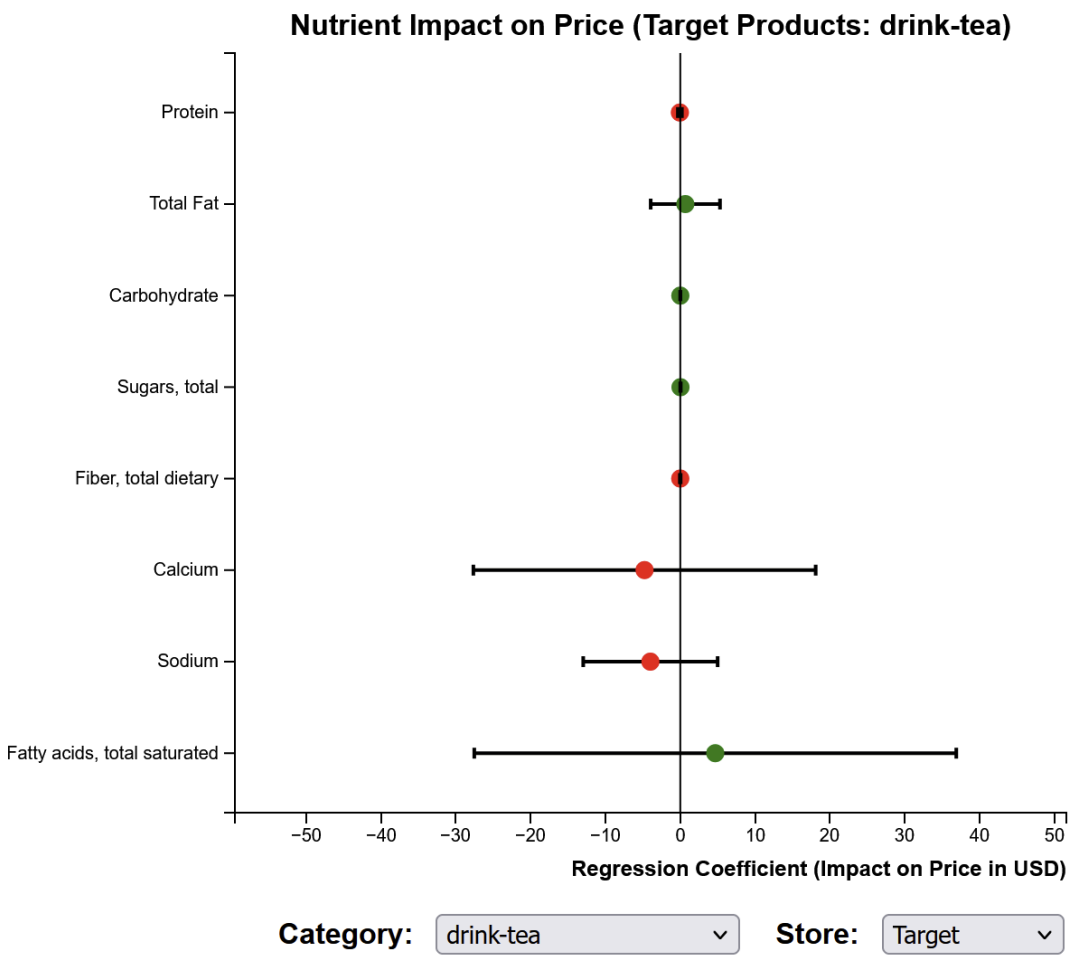
The other visualization is a coefficient plot, which would aim to estimate the potential impact of a nutrient on price (if there were an impact). The plot helps to show where there would be an increase or decrease in pricing from nutrient increases, and by around how much. Although, there are some nutrients removed from being displayed due to their data being unreliable and resulting in NaNs during calculations.



Hovering over a point pulls up a tooltip giving the exact number for the coefficient and confidence interval as it is difficult to tell just from visuals alone.



The drop down boxes allow the user to alter what data is used on the coefficient plots. The user can select what kind of product they want to analyze and from what store.



The coefficient plot also has a zoom and panning feature, when you scroll on the visual or drag it around. The zooming and panning only work horizontally. This was implemented to help users see both the larger and smaller intervals more clearly if they so desire.

Evaluation

Overall I learned from the dataset that there is likely no real impact of nutrients on the pricing of products. There is no pattern that could indicate that an increase in nutrients raises or decreases the price as the distribution of the data is usually either pretty randomly distributed without a pattern or somewhat vertical which means there is little to no change in price based on nutrient amounts.

The data also lacks points in certain categories that makes analysis unreliable as stores may not stock as many of a type of product compared to others, so the quantity and quality of the data there becomes weaker. There are also some interesting extreme cases that would cause fluctuations in the dataset too. Since there are cases where there are extreme cases in differences between nutrient quantity and pricing. For instance, having any pricing but a lack of a nutrient which was especially problematic in the cases for Vitamins which there was little data on. Another was cases like the selling of bulk products that also are already naturally packed with nutrients sold for cheap (ex: a large bag of beans) which can easily be an outlier that impacts results.

As for my specific questions that I proposed:

- Does the content or lack of content of any nutrients affect the pricings of products?
    - Majorly no, most nutrients lack any clear indication or pattern that they actually affect the pricings of products.
- Does retail at different stores affect pricing on nutrition, or is it irrelevant?
    - The product lineup does affect the pricing of products on the visuals. Different stores may shelve different products that may have wildly different pricings or nutrient contents from each other without any major clear rhythm. Additionally, the data is messed up in some stores where correlation may be inflated due to a low amount of products being stocked. An example is baby-food at Target where it may look like there is correlation from increasing prices based on protein, but there's barely any products so there aren't many points that could potentially derail the line.
- Are there any categories of products that are majorly affected by pricings based on nutrients?
    - Yes, but I would not call them conclusive or "majorly affected" as them having their pricings affected by nutrient increases. Some foods like rice-grains-packages, pudding-jello, and baby-food have some weak patterns that might indicate correlation, but them and most products overall share a general pattern of having a towering vertical cluster of points near the lower price range. This would generally imply that for around the same amount of nutrients, there can be a range of lower to higher pricings for a product.

The visuals work okay at helping readers conceptualize the general effects and potential patterns. Although, they don't help too much to create a more concrete proof of an impact. A potential improvement for the scatterplot would be to have an option to filter out outliers and draw a line between all of the current points to help convey correlation. Another potential improvement would be to increase the amount of scatterplots or coefficient plots displayed at once to create a chart of them, so you can compare multiple of them together at once.